

COMPASS_FE_20221220

Stephanie J. Wilson

2023-01-10

R Markdown

#notes I don't have a system yet to compare old slopes to current slopes for that day #I don't have a system for eliminating wells if the sd is high - if you think one of your sample wells was off you would need to remove it manually...

#Files Required: #csv of standards as list not plate with columns: Fe(II), Abs, Curve, FE ; these are: Known std concentration, Absorbance, Curve type (High or Low), and FE Curve (FE_2 or FE_Total)
#csv of samples as list not plate with columns: Date, Plate, Well_ID, SampleID, Dil_Factor, Abs1, Abs2 ; these are Date of run, Plate ID, Well ID, Sample ID, Dilution factor, Absorbance 1 (Fe2 or first read), and Absorbance 2 (feTot or second read) # I save these files in a folder for each day of FE runs

Read in Standards and subset

```
stds <- read.csv("Raw Data/FE_20230103_Stds.csv")
head(stds)
```

```
##      Fe.II.      Abs Curve      FE
## 1      500 1.3976667 High FE_Total
## 2      250 0.8980000 High FE_Total
## 3      100 0.4755000 High FE_Total
## 4       60 0.3223333 High FE_Total
## 5       45 0.2460000 High FE_Total
## 6       30 0.1846667 High FE_Total
```

```
stdid <- read.csv("Raw Data/Stds_IDs.csv")
head(stdid)
```

```
##      ID Concentration Fe_2_Low Fe_2_High Fe_3
## 1  0.5 HCL           0      Yes      Yes  Yes
## 2 Fe(II) A           60      Yes      Yes   No
## 3 Fe(II) B           45      Yes      Yes   No
## 4 Fe(II) C           30      Yes      Yes   No
## 5 Fe(II) D           18      Yes      Yes   No
## 6 Fe(II) E           9       Yes      Yes   No
```

```
stds1 <- merge(stdid, stds)
head(stds1)
```

```
##          ID Concentration Fe_2_Low Fe_2_High Fe_3 Fe.II.      Abs Curve      FE
## 1  0.5 HCL              0      Yes      Yes Yes      500 1.397667 High FE_Total
## 2 Fe(II) A              60      Yes      Yes No       500 1.397667 High FE_Total
## 3 Fe(II) B              45      Yes      Yes No       500 1.397667 High FE_Total
## 4 Fe(II) C              30      Yes      Yes No       500 1.397667 High FE_Total
## 5 Fe(II) D              18      Yes      Yes No       500 1.397667 High FE_Total
## 6 Fe(II) E              9       Yes      Yes No       500 1.397667 High FE_Total
```

```
#subset by FE_2 and FE_Total
```

```
FE_2_stds <- stds1[stds1$Fe_2_High == "Yes", ] # One factor level
head(FE_2_stds)
```

```
##          ID Concentration Fe_2_Low Fe_2_High Fe_3 Fe.II.      Abs Curve      FE
## 1  0.5 HCL              0      Yes      Yes Yes      500 1.397667 High FE_Total
## 2 Fe(II) A              60      Yes      Yes No       500 1.397667 High FE_Total
## 3 Fe(II) B              45      Yes      Yes No       500 1.397667 High FE_Total
## 4 Fe(II) C              30      Yes      Yes No       500 1.397667 High FE_Total
## 5 Fe(II) D              18      Yes      Yes No       500 1.397667 High FE_Total
## 6 Fe(II) E              9       Yes      Yes No       500 1.397667 High FE_Total
```

```
FE_Tot_stds <- stds1[stds1$Fe_3 == "Yes", ]
head(FE_Tot_stds)
```

```
# One factor level
```

```
##          ID Concentration Fe_2_Low Fe_2_High Fe_3 Fe.II.      Abs Curve
## 1  0.5 HCL              0      Yes      Yes Yes      500 1.397667 High
## 13 Fe(III) A           60      No      No  Yes      500 1.397667 High
## 14 Fe(III) B           45      No      No  Yes      500 1.397667 High
## 15 Fe(III) C           30      No      No  Yes      500 1.397667 High
## 16 Fe(III) D           18      No      No  Yes      500 1.397667 High
## 17 Fe(III) E           9       No      No  Yes      500 1.397667 High
##          FE
## 1  FE_Total
## 13 FE_Total
## 14 FE_Total
## 15 FE_Total
## 16 FE_Total
## 17 FE_Total
```

```
#subset curves for high and low
```

```
FE_2_stds_high <- FE_2_stds[FE_2_stds$Fe_2_High == "Yes", ]
head(FE_2_stds_high)
```

```
# One factor
```

```
##          ID Concentration Fe_2_Low Fe_2_High Fe_3 Fe.II.      Abs Curve      FE
## 1  0.5 HCL              0      Yes      Yes Yes      500 1.397667 High FE_Total
## 2 Fe(II) A              60      Yes      Yes No       500 1.397667 High FE_Total
## 3 Fe(II) B              45      Yes      Yes No       500 1.397667 High FE_Total
## 4 Fe(II) C              30      Yes      Yes No       500 1.397667 High FE_Total
## 5 Fe(II) D              18      Yes      Yes No       500 1.397667 High FE_Total
## 6 Fe(II) E              9       Yes      Yes No       500 1.397667 High FE_Total
```

```
FE_2_stds_low <- FE_2_stds[FE_2_stds$Fe_2_Low == "Yes", ] # One factor l
head(FE_2_stds_low)
```

```
##      ID Concentration Fe_2_Low Fe_2_High Fe_3 Fe.II.      Abs Curve      FE
## 1  0.5 HCL           0      Yes      Yes  Yes    500 1.397667 High FE_Total
## 2 Fe(II) A           60      Yes      Yes   No    500 1.397667 High FE_Total
## 3 Fe(II) B           45      Yes      Yes   No    500 1.397667 High FE_Total
## 4 Fe(II) C           30      Yes      Yes   No    500 1.397667 High FE_Total
## 5 Fe(II) D           18      Yes      Yes   No    500 1.397667 High FE_Total
## 6 Fe(II) E            9      Yes      Yes   No    500 1.397667 High FE_Total
```

```
FE_Tot_stds_high <- FE_Tot_stds[FE_Tot_stds$Curve == "High", ] # One fac
head(FE_Tot_stds_high)
```

```
##      ID Concentration Fe_2_Low Fe_2_High Fe_3 Fe.II.      Abs Curve
## 1  0.5 HCL           0      Yes      Yes  Yes    500 1.397667 High
## 13 Fe(III) A         60      No      No   Yes    500 1.397667 High
## 14 Fe(III) B         45      No      No   Yes    500 1.397667 High
## 15 Fe(III) C         30      No      No   Yes    500 1.397667 High
## 16 Fe(III) D         18      No      No   Yes    500 1.397667 High
## 17 Fe(III) E          9      No      No   Yes    500 1.397667 High
##      FE
## 1  FE_Total
## 13 FE_Total
## 14 FE_Total
## 15 FE_Total
## 16 FE_Total
## 17 FE_Total
```

```
FE_Tot_stds_low <- FE_Tot_stds[FE_Tot_stds$Curve == "Low", ] # One facto
head(FE_Tot_stds_low)
```

```
##      ID Concentration Fe_2_Low Fe_2_High Fe_3 Fe.II.      Abs Curve
## 529 0.5 HCL           0      Yes      Yes  Yes    60 0.3223333 Low
## 541 Fe(III) A         60      No      No   Yes    60 0.3223333 Low
## 542 Fe(III) B         45      No      No   Yes    60 0.3223333 Low
## 543 Fe(III) C         30      No      No   Yes    60 0.3223333 Low
## 544 Fe(III) D         18      No      No   Yes    60 0.3223333 Low
## 545 Fe(III) E          9      No      No   Yes    60 0.3223333 Low
##      FE
## 529 FE_Total
## 541 FE_Total
## 542 FE_Total
## 543 FE_Total
## 544 FE_Total
## 545 FE_Total
```

Plot standards

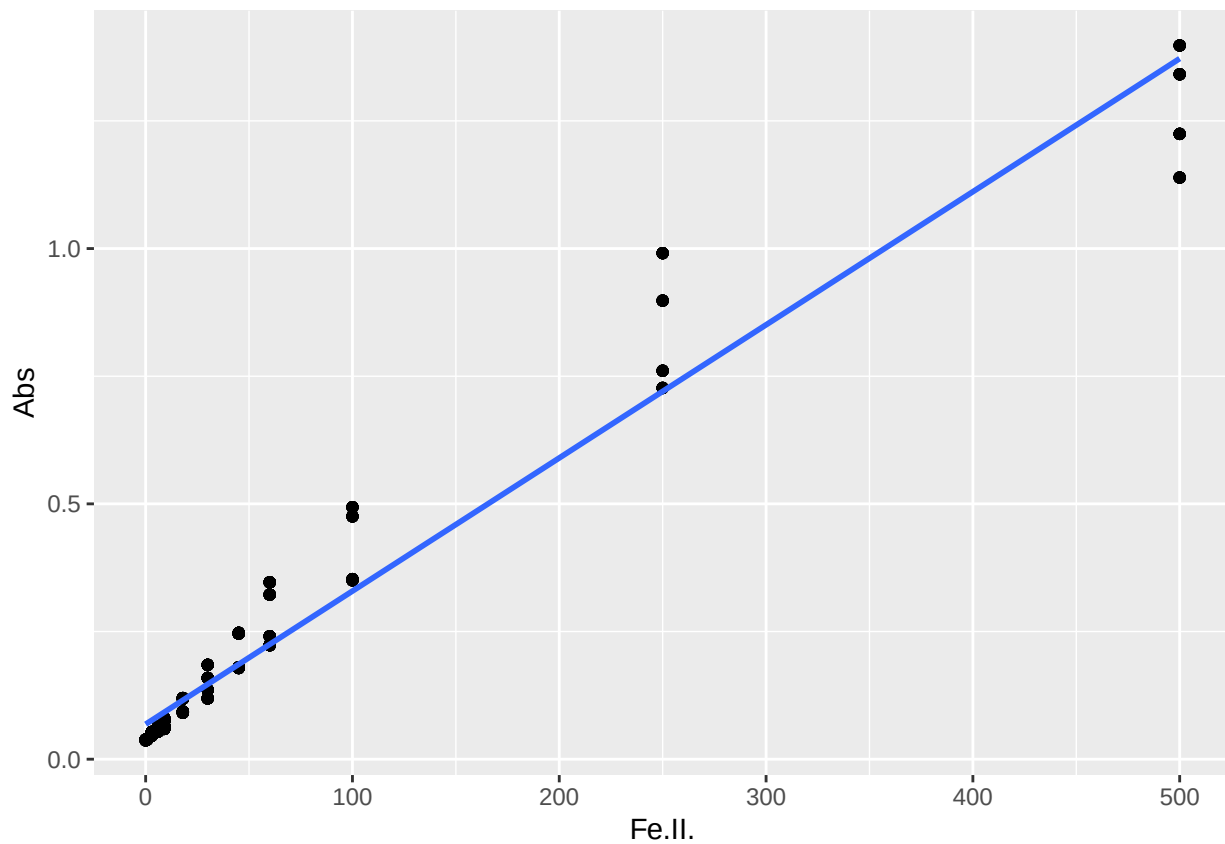
```
#Plot each of these and calculate the slope, intercept, and R2
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```
#Low curve for Fe(II)
```

```
FE_2_Low <- ggplot(FE_2_stds_low, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se = FALSE)
FE_2_Low
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_2_Low_lm <- lm(FE_2_stds_low$Abs ~ FE_2_stds_low$Fe.II.)
summary(FE_2_Low_lm)
```

```
##
## Call:
## lm(formula = FE_2_stds_low$Abs ~ FE_2_stds_low$Fe.II.)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23302 -0.03036 -0.02050  0.01227  0.27073
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.852e-02  2.577e-03   26.59  <2e-16 ***
## FE_2_stds_low$Fe.II. 2.607e-03  2.036e-05  128.07  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06331 on 754 degrees of freedom
## Multiple R-squared:  0.956, Adjusted R-squared:  0.956
## F-statistic: 1.64e+04 on 1 and 754 DF, p-value: < 2.2e-16
```

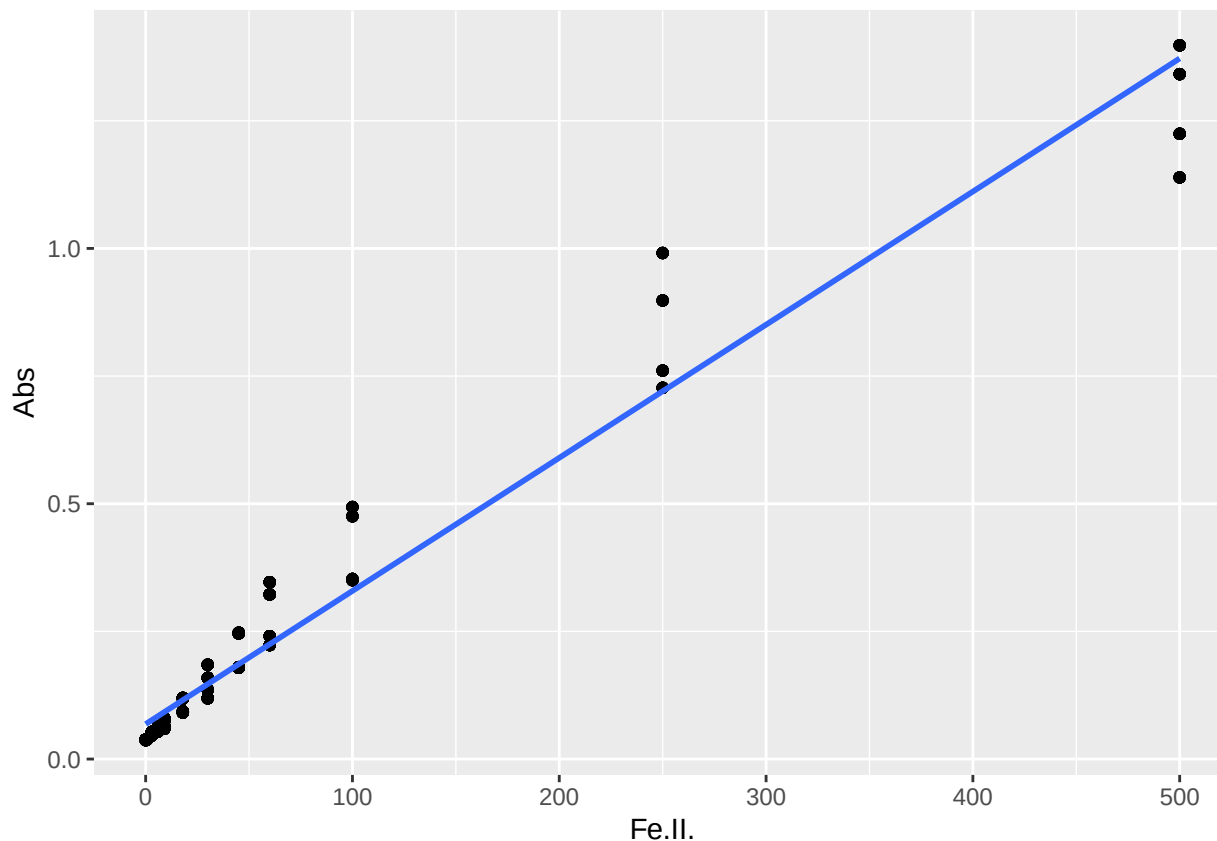
```
cf <- coef(FE_2_Low_lm)

#create data frame with 1 rows and 0 columns
Slopes1 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes1$Curve <- "FE_2_Low"
Slopes1$R2 <- summary(FE_2_Low_lm)$adj.r.squared
Slopes1$Slope <- cf[2]
Slopes1$Intercept <- cf[1]
head(Slopes1)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_2_Low 0.9559893 0.002606997 0.06852172
```

```
#High curve for Fe(III)
FE_2_High <- ggplot(FE_2_stds_high, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se = 1)
FE_2_High
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_2_High_lm <- lm(FE_2_stds_high$Abs ~ FE_2_stds_high$Fe.II.)
summary(FE_2_High_lm)
```

```
##
## Call:
## lm(formula = FE_2_stds_high$Abs ~ FE_2_stds_high$Fe.II.)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23302 -0.03036 -0.02050  0.01227  0.27073
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.852e-02  2.231e-03  30.72  <2e-16 ***
## FE_2_stds_high$Fe.II. 2.607e-03  1.762e-05 147.93  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06329 on 1006 degrees of freedom
## Multiple R-squared:  0.956, Adjusted R-squared:  0.956
## F-statistic: 2.188e+04 on 1 and 1006 DF, p-value: < 2.2e-16
```

```
cf <- coef(FE_2_High_lm)
```

```
#create data frame with 1 rows and 0 columns
```

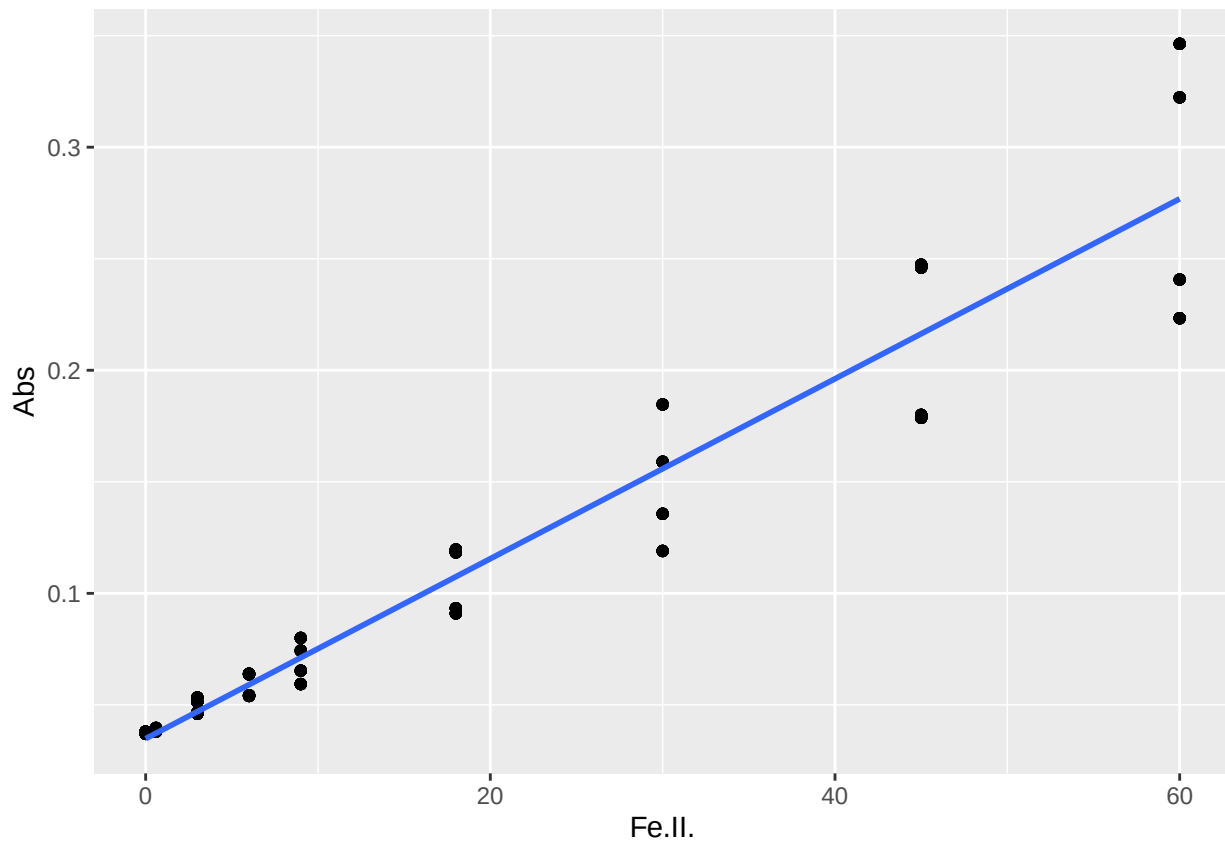
```
Slopes2 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes2$Curve <- "FE_2_High"
Slopes2$R2 <- summary(FE_2_High_lm)$adj.r.squared
Slopes2$Slope <- cf[2]
Slopes2$Intercept <- cf[1]
head(Slopes2)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_2_High 0.9560039 0.002606997 0.06852172
```

```
#Low curve for Fe(Total)
```

```
FE_Tot_Low <- ggplot(FE_Tot_stds_low, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se = FALSE)
FE_Tot_Low
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_Tot_Low_lm <- lm(FE_Tot_stds_low$Abs ~ FE_Tot_stds_low$Fe.II.)
summary(FE_Tot_Low_lm)
```

```
##
## Call:
## lm(formula = FE_Tot_stds_low$Abs ~ FE_Tot_stds_low$Fe.II.)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.053522 -0.007336  0.002252  0.005288  0.069478
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.487e-02  1.607e-03   21.70  <2e-16 ***
## FE_Tot_stds_low$Fe.II. 4.033e-03  5.773e-05   69.86  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02331 on 394 degrees of freedom
## Multiple R-squared:  0.9253, Adjusted R-squared:  0.9251
## F-statistic: 4880 on 1 and 394 DF, p-value: < 2.2e-16
```

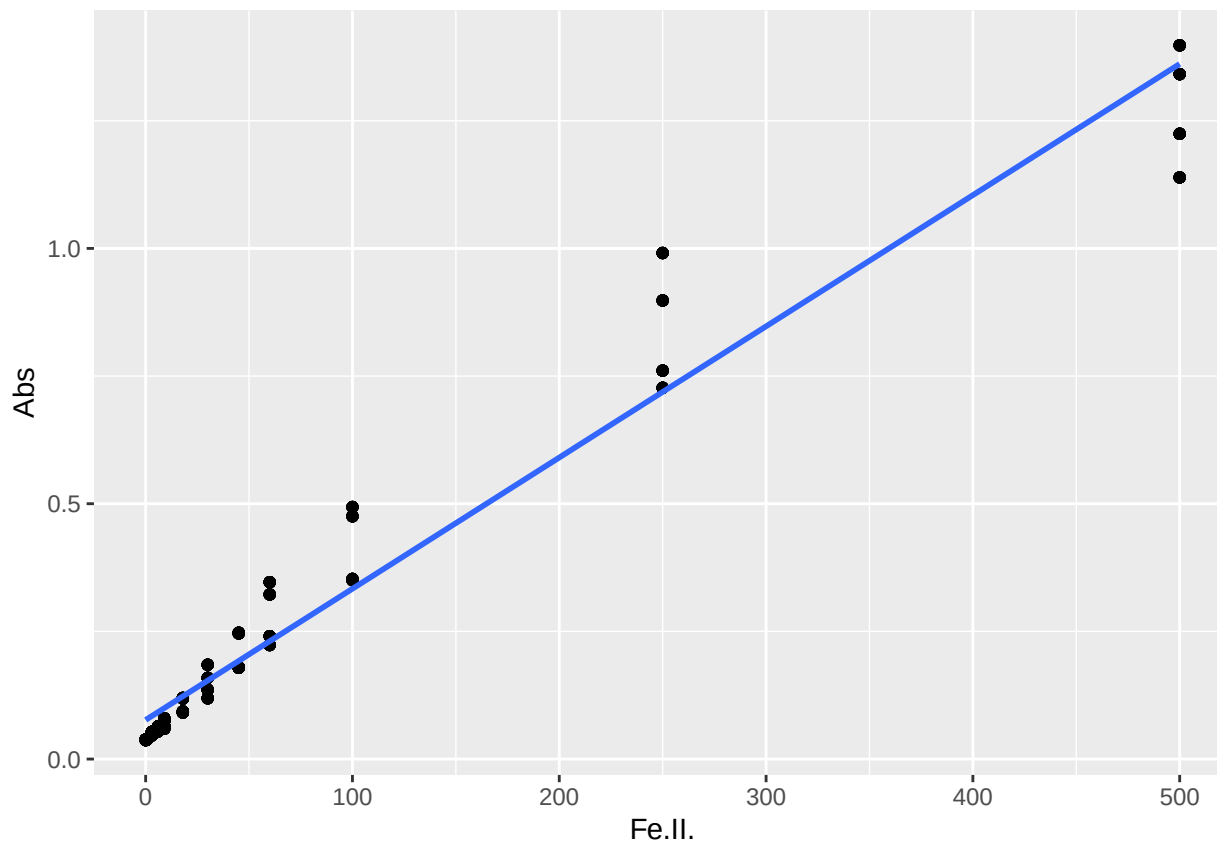
```
cf <- coef(FE_Tot_Low_lm)

#create data frame with 1 rows and 0 columns
Slopes3 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes3$Curve <- "FE_Tot_Low"
Slopes3$R2 <- summary(FE_Tot_Low_lm)$adj.r.squared
Slopes3$Slope <- cf[2]
Slopes3$Intercept <- cf[1]
head(Slopes3)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_Tot_Low 0.9251034 0.004033061 0.03487149
```

```
#High curve for Fe(III)
FE_Tot_High <- ggplot(FE_Tot_stds_high, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", s
FE_Tot_High
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_Tot_High_lm <- lm(FE_Tot_stds_high$Abs ~ FE_Tot_stds_high$Fe.II.)
summary(FE_Tot_High_lm)
```

```
##
## Call:
## lm(formula = FE_Tot_stds_high$Abs ~ FE_Tot_stds_high$Fe.II.)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.22226 -0.03806 -0.02264  0.01741  0.27209
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.657e-02  3.897e-03   19.65  <2e-16 ***
## FE_Tot_stds_high$Fe.II. 2.569e-03  2.352e-05  109.26  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07682 on 526 degrees of freedom
## Multiple R-squared:  0.9578, Adjusted R-squared:  0.9577
## F-statistic: 1.194e+04 on 1 and 526 DF, p-value: < 2.2e-16
```

```
cf <- coef(FE_Tot_High_lm)
```

```
#create data frame with 1 rows and 0 columns
```

```
Slopes4 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes4$Curve <- "FE_Tot_High"
Slopes4$R2 <- summary(FE_Tot_High_lm)$adj.r.squared
Slopes4$Slope <- cf[2]
Slopes4$Intercept <- cf[1]
head(Slopes4)
```

```
##           Curve           R2           Slope Intercept
## 1 FE_Tot_High 0.9577141 0.002569379 0.07656924
```

```
#pull all the slopes together into one dataframe
Slopes <- rbind(Slopes1, Slopes2, Slopes3, Slopes4)

####
```

Read in Sample Data subset

```
dat <- read.csv("Raw Data/FE_20230103_Samples.csv")
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230103 Samples1         1      stds              1 1.138 1.315
## 2 20230103 Samples1         2      stds              1 1.312 1.401    BR
## 3 20230103 Samples1         3      stds              1 1.188 1.472
## 4 20230103 Samples1         4        270              1 0.055 0.040    ND
## 5 20230103 Samples1         5        270              1 0.051 0.039    ND
## 6 20230103 Samples1         6        270              1 0.053 0.041    ND
```

```
## remove any lines that say stds in SampleID
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##      filter, lag

## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
```

```
dat <- dat %>% filter(!SampleID == 'stds')
dat <- dat %>% filter(!Notes == 'BR')
dat <- dat %>% filter(!Notes == 'ND')
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230103 Samples1      7      278          1 0.066 0.094
## 2 20230103 Samples1      9      278          1 0.074 0.100
## 3 20230103 Samples1     10      284          7 0.048 0.052
## 4 20230103 Samples1     11      284          7 0.048 0.053
## 5 20230103 Samples1     12      284          7 0.048 0.055
## 6 20230103 Samples1     19      279          7 0.194 0.292
```

```
#Now flag any Abs1 that are above the 60uM standard absorbance (0.2)
dat$FE_2_Curve <- ifelse(dat$Abs1 >0.2, "High", "Low")
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230103 Samples1      7      278          1 0.066 0.094
## 2 20230103 Samples1      9      278          1 0.074 0.100
## 3 20230103 Samples1     10      284          7 0.048 0.052
## 4 20230103 Samples1     11      284          7 0.048 0.053
## 5 20230103 Samples1     12      284          7 0.048 0.055
## 6 20230103 Samples1     19      279          7 0.194 0.292
##   FE_2_Curve
## 1         Low
## 2         Low
## 3         Low
## 4         Low
## 5         Low
## 6         Low
```

```
#Now flag any Abs2 that are above the 60uM standard absorbance (0.3)
dat$FE_Tot_Curve <- ifelse(dat$Abs2 >0.3, "High", "Low")
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230103 Samples1      7      278          1 0.066 0.094
## 2 20230103 Samples1      9      278          1 0.074 0.100
## 3 20230103 Samples1     10      284          7 0.048 0.052
## 4 20230103 Samples1     11      284          7 0.048 0.053
## 5 20230103 Samples1     12      284          7 0.048 0.055
## 6 20230103 Samples1     19      279          7 0.194 0.292
##   FE_2_Curve FE_Tot_Curve
## 1         Low         Low
## 2         Low         Low
## 3         Low         Low
## 4         Low         Low
## 5         Low         Low
## 6         Low         Low
```

Subset Data and Calculate Concentrations

```
#Calculate concentrations of Fe(II)
dat$FE_2_Conc <- ifelse(dat$Abs1 >0.2, (dat$Abs1-Slopes2$Intercept)/Slopes2$Slope, (dat$Abs1-Slopes1$In
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230103 Samples1      7      278      1 0.066 0.094
## 2 20230103 Samples1      9      278      1 0.074 0.100
## 3 20230103 Samples1     10      284      7 0.048 0.052
## 4 20230103 Samples1     11      284      7 0.048 0.053
## 5 20230103 Samples1     12      284      7 0.048 0.055
## 6 20230103 Samples1     19      279      7 0.194 0.292
##   FE_2_Curve FE_Tot_Curve FE_2_Conc
## 1      Low      Low -0.9672878
## 2      Low      Low  2.1013769
## 3      Low      Low -7.8717834
## 4      Low      Low -7.8717834
## 5      Low      Low -7.8717834
## 6      Low      Low 48.1313473
```

```
#calculate concentrations of Fe(total)
```

```
dat$FE_Tot_Conc <- ifelse(dat$Abs2 > 0.3, (dat$Abs2-Slopes4$Intercept)/Slopes4$Slope,
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230103 Samples1      7      278      1 0.066 0.094
## 2 20230103 Samples1      9      278      1 0.074 0.100
## 3 20230103 Samples1     10      284      7 0.048 0.052
## 4 20230103 Samples1     11      284      7 0.048 0.053
## 5 20230103 Samples1     12      284      7 0.048 0.055
## 6 20230103 Samples1     19      279      7 0.194 0.292
##   FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc
## 1      Low      Low -0.9672878 14.660951
## 2      Low      Low  2.1013769 16.148655
## 3      Low      Low -7.8717834  4.247025
## 4      Low      Low -7.8717834  4.494976
## 5      Low      Low -7.8717834  4.990877
## 6      Low      Low 48.1313473 63.755175
```

```
#Account for the dilution factor
```

```
dat$FE_2_Conc_DilCorr <- dat$FE_2_Conc*dat$dilution.factor
```

```
dat$FE_Tot_Conc_DilCorr <- dat$FE_Tot_Conc*dat$dilution.factor
```

```
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230103 Samples1      7      278      1 0.066 0.094
## 2 20230103 Samples1      9      278      1 0.074 0.100
## 3 20230103 Samples1     10      284      7 0.048 0.052
## 4 20230103 Samples1     11      284      7 0.048 0.053
## 5 20230103 Samples1     12      284      7 0.048 0.055
## 6 20230103 Samples1     19      279      7 0.194 0.292
##   FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1      Low      Low -0.9672878 14.660951 -0.9672878
## 2      Low      Low  2.1013769 16.148655  2.1013769
## 3      Low      Low -7.8717834  4.247025 -55.1024839
## 4      Low      Low -7.8717834  4.494976 -55.1024839
```

```
## 5      Low      Low -7.8717834    4.990877    -55.1024839
## 6      Low      Low 48.1313473    63.755175    336.9194314
##  FE_Tot_Conc_DilCorr
## 1          14.66095
## 2          16.14866
## 3          29.72918
## 4          31.46483
## 5          34.93614
## 6          446.28622
```

Calculate Fe(III)

```
#Calculate concentrations of Fe(III)
dat$FE_3_Conc <- dat$FE_Tot_Conc_DilCorr-dat$FE_2_Conc_DilCorr

#Use ifelse to make any negative values equal to zero
dat$FE_2_Conc_Final <- ifelse(dat$FE_2_Conc_DilCorr <0, 0, dat$FE_2_Conc_DilCorr)

dat$FE_3_Conc_Final <- ifelse(dat$FE_3_Conc <0, 0, dat$FE_3_Conc)

#Use ifelse to make any negative values equal to zero
dat$FE_2_Info <- ifelse(dat$FE_2_Conc_DilCorr <0, 'bd1', 'within range')

#Use ifelse to make any negative values equal to zero
dat$FE_3_Info <- ifelse(dat$FE_3_Conc <0, 'bd1', 'within range')

head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor  Abs1  Abs2 Notes
## 1 20230103 Samples1      7      278      1 0.066 0.094
## 2 20230103 Samples1      9      278      1 0.074 0.100
## 3 20230103 Samples1     10      284      7 0.048 0.052
## 4 20230103 Samples1     11      284      7 0.048 0.053
## 5 20230103 Samples1     12      284      7 0.048 0.055
## 6 20230103 Samples1     19      279      7 0.194 0.292
##  FE_2_Curve FE_Tot_Curve  FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1      Low      Low -0.9672878    14.660951    -0.9672878
## 2      Low      Low  2.1013769    16.148655     2.1013769
## 3      Low      Low -7.8717834    4.247025    -55.1024839
## 4      Low      Low -7.8717834    4.494976    -55.1024839
## 5      Low      Low -7.8717834    4.990877    -55.1024839
## 6      Low      Low 48.1313473    63.755175    336.9194314
##  FE_Tot_Conc_DilCorr FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final  FE_2_Info
## 1          14.66095  15.62824      0.000000      15.62824      bd1
## 2          16.14866  14.04728      2.101377      14.04728 within range
## 3          29.72918  84.83166      0.000000      84.83166      bd1
## 4          31.46483  86.56731      0.000000      86.56731      bd1
## 5          34.93614  90.03862      0.000000      90.03862      bd1
## 6          446.28622 109.36679      336.919431    109.36679 within range
##      FE_3_Info
## 1 within range
## 2 within range
```

```
## 3 within range
## 4 within range
## 5 within range
## 6 within range
```

Calculate Averages across wells and std. dev.

```
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230103 Samples1         7      278          1 0.066 0.094
## 2 20230103 Samples1         9      278          1 0.074 0.100
## 3 20230103 Samples1        10      284          7 0.048 0.052
## 4 20230103 Samples1        11      284          7 0.048 0.053
## 5 20230103 Samples1        12      284          7 0.048 0.055
## 6 20230103 Samples1        19      279          7 0.194 0.292
##      FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1          Low          Low -0.9672878  14.660951      -0.9672878
## 2          Low          Low  2.1013769  16.148655       2.1013769
## 3          Low          Low -7.8717834   4.247025     -55.1024839
## 4          Low          Low -7.8717834   4.494976     -55.1024839
## 5          Low          Low -7.8717834   4.990877     -55.1024839
## 6          Low          Low 48.1313473  63.755175    336.9194314
##      FE_Tot_Conc_DilCorr FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final FE_2_Info
## 1          14.66095  15.62824      0.000000      15.62824      bdl
## 2          16.14866  14.04728      2.101377      14.04728 within range
## 3          29.72918  84.83166      0.000000      84.83166      bdl
## 4          31.46483  86.56731      0.000000      86.56731      bdl
## 5          34.93614  90.03862      0.000000      90.03862      bdl
## 6          446.28622 109.36679      336.919431    109.36679 within range
##      FE_3_Info
## 1 within range
## 2 within range
## 3 within range
## 4 within range
## 5 within range
## 6 within range
```

```
#summarize by sampleID so that we can calculate the mean and std. dev. of the three wells
dat1 <- dat %>%
```

```
  group_by(SampleID) %>%
  summarise(FE_2_mean = mean(FE_2_Conc_Final), FE_2_sd = sd(FE_2_Conc_Final),
            FE_3_mean = mean(FE_3_Conc_Final), FE_3_sd = sd(FE_3_Conc_Final),
            FE_2_flag = first(FE_2_Info), FE_3_flag = first(FE_3_Info))
```

```
head(dat1)
```

```
## # A tibble: 6 x 7
##   SampleID FE_2_mean FE_2_sd FE_3_mean FE_3_sd FE_2_flag FE_3_flag
##   <chr>      <dbl> <dbl> <dbl> <dbl> <chr>      <chr>
## 1 266      198.    6.24  110.  10.8 within range within range
```

```
## 2 267      450.      12.7      43.1      31.7  within range within range
## 3 268      516.       7.87      50.5       2.04 within range within range
## 4 269      502.      55.7      108.      51.1  within range within range
## 5 278        1.05      1.49      14.8      1.12 bdl          within range
## 6 279      352.      20.9      92.0      24.6  within range within range
```

Export full data then just final data

```
#Read in meta-data and sample IDs
IDs <- read.csv("Raw Data/FE_SampleIDs_2022.csv")
head(IDs)
```

```
##      Date Month Year Campaign Site Zone Depth Rep
## 1 20220502   May 2022   MonMon  MSM   WC    10   A
## 2 20220502   May 2022   MonMon  MSM   WC    10   B
## 3 20220502   May 2022   MonMon  MSM   WC    10   C
## 4 20220502   May 2022   MonMon  MSM   WC    20   A
## 5 20220502   May 2022   MonMon  MSM   WC    20   B
## 6 20220502   May 2022   MonMon  MSM   WC    20   C
##
##      Sample_Name SampleID
## 1 MSM_WC_PW_MonMon_20220502_SipA_10cm      654
## 2 MSM_WC_PW_MonMon_20220502_SipB_10cm      656
## 3 MSM_WC_PW_MonMon_20220502_SipC_10cm      658
## 4 MSM_WC_PW_MonMon_20220502_SipA_20cm      655
## 5 MSM_WC_PW_MonMon_20220502_SipB_20cm      657
## 6 MSM_WC_PW_MonMon_20220502_SipC_20cm      659
```

```
dat_IDs <- merge(IDs, dat1)
head(dat_IDs)
```

```
##      SampleID   Date      Month Year Campaign Site Zone Depth Rep
## 1      266 202210   October 2022   MonMon  MSM   WC     1   1
## 2      267 202210   October 2022   MonMon  MSM   WC     2   2
## 3      268 202210   October 2022   MonMon  MSM   WC     3   3
## 4      269 202210   October 2022   MonMon  MSM   WC     4   4
## 5      278 202209 September 2022   MonMon Gcrew  UP    20   A
## 6      279 202209 September 2022   MonMon Gcrew  TR    20   A
##
##      Sample_Name FE_2_mean  FE_2_sd FE_3_mean  FE_3_sd  FE_2_flag
## 1 MSM_202210_WC_PEEPER_1 197.920543  6.238387 110.49281 10.824344 within range
## 2 MSM_202210_WC_PEEPER_2 449.551048 12.748008  43.14809 31.737176 within range
## 3 MSM_202210_WC_PEEPER_3 516.294506  7.865792  50.54707  2.041600 within range
## 4 MSM_202210_WC_PEEPER_4 501.526557 55.651283 108.38638 51.081814 within range
## 5   GCrew_UP_LysA_20cm   1.050688  1.485898  14.83776  1.117908      bdl
## 6   GCrew_TR_LysA_20cm 351.687380 20.885034  91.99536 24.566912 within range
##
##      FE_3_flag
## 1 within range
## 2 within range
## 3 within range
## 4 within range
## 5 within range
## 6 within range
```

```
#Read out the summarized data
head(dat_IDs)
```

```
##      SampleID   Date      Month Year Campaign Site Zone Depth Rep
## 1      266 202210   October 2022   MonMon   MSM   WC      1    1
## 2      267 202210   October 2022   MonMon   MSM   WC      2    2
## 3      268 202210   October 2022   MonMon   MSM   WC      3    3
## 4      269 202210   October 2022   MonMon   MSM   WC      4    4
## 5      278 202209   September 2022   MonMon Gcrew   UP     20    A
## 6      279 202209   September 2022   MonMon Gcrew   TR     20    A
##      Sample_Name FE_2_mean FE_2_sd FE_3_mean FE_3_sd FE_2_flag
## 1 MSM_202210_WC_PEEPER_1 197.920543 6.238387 110.49281 10.824344 within range
## 2 MSM_202210_WC_PEEPER_2 449.551048 12.748008 43.14809 31.737176 within range
## 3 MSM_202210_WC_PEEPER_3 516.294506 7.865792 50.54707 2.041600 within range
## 4 MSM_202210_WC_PEEPER_4 501.526557 55.651283 108.38638 51.081814 within range
## 5 GCrew_UP_LysA_20cm 1.050688 1.485898 14.83776 1.117908 bdl
## 6 GCrew_TR_LysA_20cm 351.687380 20.885034 91.99536 24.566912 within range
##      FE_3_flag
## 1 within range
## 2 within range
## 3 within range
## 4 within range
## 5 within range
## 6 within range
```

```
write.csv(dat_IDs, file="Processed Data/FE_20230103_Summary_Data.csv")
```

```
#Read out all the data in dat
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230103 Samples1      7      278      1 0.066 0.094
## 2 20230103 Samples1      9      278      1 0.074 0.100
## 3 20230103 Samples1     10      284      7 0.048 0.052
## 4 20230103 Samples1     11      284      7 0.048 0.053
## 5 20230103 Samples1     12      284      7 0.048 0.055
## 6 20230103 Samples1     19      279      7 0.194 0.292
##      FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1      Low      Low -0.9672878 14.660951 -0.9672878
## 2      Low      Low 2.1013769 16.148655 2.1013769
## 3      Low      Low -7.8717834 4.247025 -55.1024839
## 4      Low      Low -7.8717834 4.494976 -55.1024839
## 5      Low      Low -7.8717834 4.990877 -55.1024839
## 6      Low      Low 48.1313473 63.755175 336.9194314
##      FE_Tot_Conc_DilCorr FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final FE_2_Info
## 1      14.66095 15.62824 0.000000 15.62824 bdl
## 2      16.14866 14.04728 2.101377 14.04728 within range
## 3      29.72918 84.83166 0.000000 84.83166 bdl
## 4      31.46483 86.56731 0.000000 86.56731 bdl
## 5      34.93614 90.03862 0.000000 90.03862 bdl
## 6      446.28622 109.36679 336.919431 109.36679 within range
##      FE_3_Info
## 1 within range
```

```
## 2 within range
## 3 within range
## 4 within range
## 5 within range
## 6 within range
```

```
write.csv(dat, file="Processed Data/FE_20230103_Data.csv")
```

```
#Now take out the absorbance and stuff
head(dat)
```

```
##      Date      Plate Well.Number SampleID dilution.factor Abs1 Abs2 Notes
## 1 20230103 Samples1      7      278          1 0.066 0.094
## 2 20230103 Samples1      9      278          1 0.074 0.100
## 3 20230103 Samples1     10      284          7 0.048 0.052
## 4 20230103 Samples1     11      284          7 0.048 0.053
## 5 20230103 Samples1     12      284          7 0.048 0.055
## 6 20230103 Samples1     19      279          7 0.194 0.292
##  FE_2_Curve FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr
## 1      Low      Low -0.9672878  14.660951    -0.9672878
## 2      Low      Low  2.1013769  16.148655     2.1013769
## 3      Low      Low -7.8717834   4.247025    -55.1024839
## 4      Low      Low -7.8717834   4.494976    -55.1024839
## 5      Low      Low -7.8717834   4.990877    -55.1024839
## 6      Low      Low 48.1313473  63.755175    336.9194314
##  FE_Tot_Conc_DilCorr FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final FE_2_Info
## 1      14.66095  15.62824      0.000000      15.62824      bdl
## 2      16.14866  14.04728      2.101377      14.04728 within range
## 3      29.72918  84.83166      0.000000      84.83166      bdl
## 4      31.46483  86.56731      0.000000      86.56731      bdl
## 5      34.93614  90.03862      0.000000      90.03862      bdl
## 6      446.28622 109.36679      336.919431 109.36679 within range
##      FE_3_Info
## 1 within range
## 2 within range
## 3 within range
## 4 within range
## 5 within range
## 6 within range
```

```
dat2 <- dat[, -(2:3)]
dat2 <- dat2[, -(3:12)]
head(dat2)
```

```
##      Date SampleID FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final FE_2_Info
## 1 20230103      278  15.62824      0.000000      15.62824      bdl
## 2 20230103      278  14.04728      2.101377      14.04728 within range
## 3 20230103      284  84.83166      0.000000      84.83166      bdl
## 4 20230103      284  86.56731      0.000000      86.56731      bdl
## 5 20230103      284  90.03862      0.000000      90.03862      bdl
## 6 20230103      279 109.36679      336.919431 109.36679 within range
##      FE_3_Info
```

```
## 1 within range  
## 2 within range  
## 3 within range  
## 4 within range  
## 5 within range  
## 6 within range
```

```
write.csv(dat2, file="Processed Data/FE_20230103_short_Data.csv")
```

END